

ABSTRACT

Wireless sensor networks deployed in large or obstructed environments often struggle with maintaining continuous connectivity due to limited communication range, terrain constraints, and power limitations. These challenges affect the reliability of data collection, especially in applications that rely on distributed sensing. To address this issue, our work explores a mobile, autonomous approach for aggregating data from spatially separated sensor nodes using short-range wireless communication. A compact robotic platform was developed to navigate predefined regions, interact with sensor nodes, and transfer collected information to a centralized monitoring unit. Low-power communication techniques were incorporated to ensure energy-efficient operation for both mobile and stationary components. The system was evaluated under multiple environmental and operational conditions to observe node detection performance, data transfer stability, and mobility behaviour.

The study highlights how autonomous mobility can complement traditional IoT networks by overcoming connectivity gaps, reducing power burden on sensor nodes, and improving overall data reliability. The findings open pathways for future enhancements in navigation intelligence, long-range communication integration, and scalable deployment for agriculture, environmental monitoring, and other distributed sensing applications.

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INTRODUCTION

In many real-world environments, continuous wireless connectivity is not always possible due to long distances, obstacles, or limited power availability. To address this challenge, the concept of a *Data Mule* offers an efficient solution by using a mobile device to collect data from distributed sensor nodes.

The MuleBot project implements this idea using an ESP32-based mobile robot that autonomously travels within a defined area, detects nearby sensor nodes, and securely collects their data using short-range ESP-NOW communication. After gathering information, the MuleBot delivers the collected data to a base station in a structured format.

This system provides an inexpensive, energy-efficient, and reliable method for data aggregation in scenarios such as agriculture, environmental monitoring, smart campuses, restricted zones, and defence applications. The project demonstrates how low-cost embedded systems and IoT technologies can be combined to solve connectivity challenges in remote sensor networks.

PROBLEM STATEMENT

In remote, large-scale, or obstructed environments, wireless sensor nodes often face challenges such as limited communication range, poor network coverage, and high energy consumption. As a result, sensor data cannot be reliably transferred to a central base station. Traditional solutions like long-range communication modules are costly and drain battery power quickly.

There is a need for a low-cost, energy-efficient, and reliable method to collect data from distributed sensor nodes without requiring continuous network connectivity.

OBJECTIVES

1. To design and develop a mobile MuleBot capable of autonomously navigating within a defined region.
2. To implement ESP-NOW–based communication for low-power, short-range, and secure data transfer between the MuleBot and sensor nodes.
3. To detect sensor nodes automatically and collect environmental or application-specific data from each node.
4. To store and forward data to a base station in a structured (CSV) format.
5. To create a low-cost, energy-efficient solution suitable for agriculture, environmental monitoring, and remote sensing applications.

METHODOLOGY

1. System Requirements and Design

The functional requirements were identified, including autonomous movement, node detection, data collection, and data transfer. A block diagram was designed showing the interaction between the MuleBot, sensor nodes, and the base station.

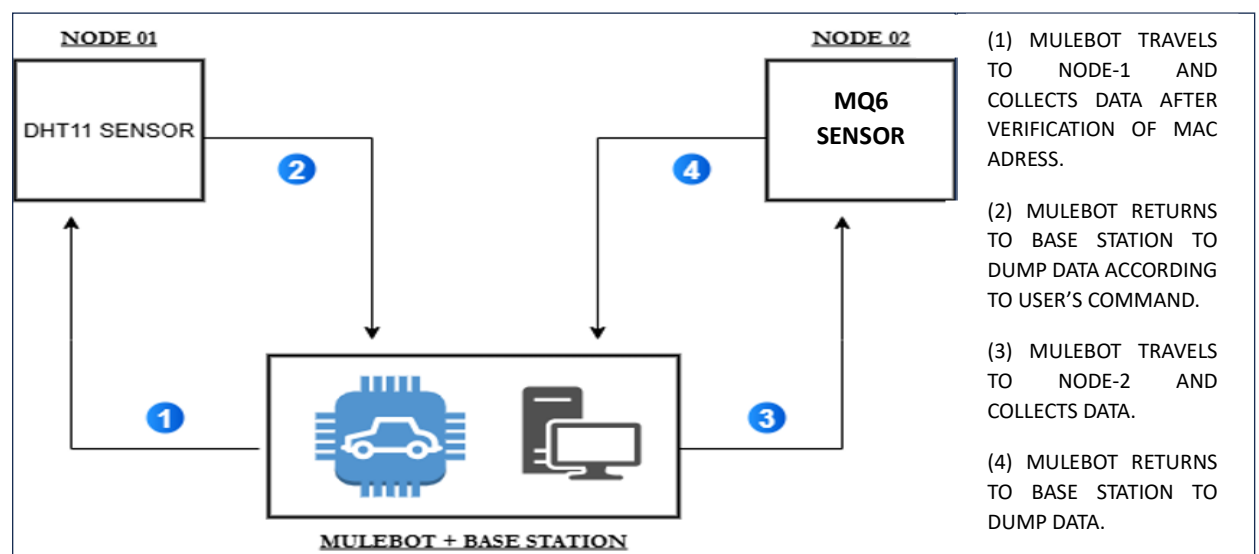


Figure 1. Block Diagram of the system

2. Hardware Development

The MuleBot was built using an ESP32 microcontroller, motor driver, DC motors, battery pack, and onboard sensors. Each sensor node was designed using an ESP32 with environmental sensors such as temperature, humidity, or soil moisture.



Figure 2. Mule bot

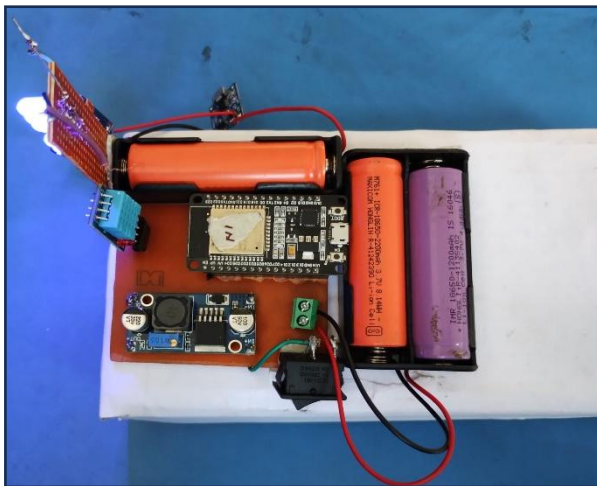


Figure 3. Top view of Node 1

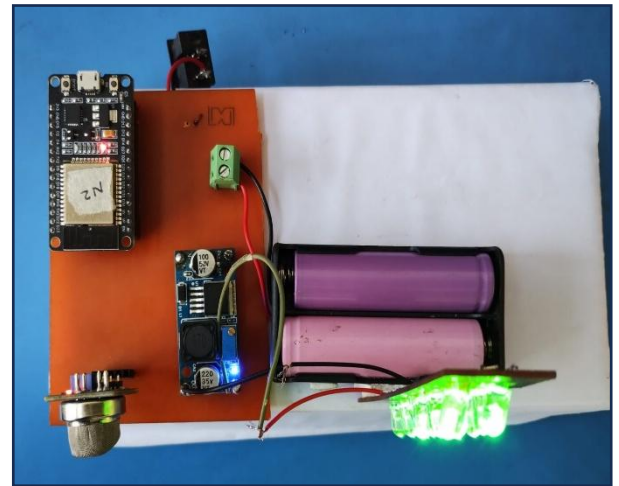


Figure 4. Top view of Node 2

3. Communication Setup using ESP-NOW

ESP-NOW communication protocol was implemented due to its low power consumption, high reliability, and ability to function without Wi-Fi. Unique MAC addresses were assigned for secure pairing between the MuleBot and sensor nodes.

4. Autonomous Navigation Implementation

Algorithms were developed for controlled movement of the MuleBot. Basic waypoint navigation or obstacle-avoidance logic was incorporated to ensure the robot moves through the target area.

5. Node Detection and Data Collection

The MuleBot continuously scanned for registered sensor nodes. When a node was detected, the MuleBot initiated an ESP-NOW data request and collected the transmitted sensor data securely.

6. Data Storage and Processing

Received data was stored temporarily in the memory of MuleBot. The data was formatted into CSV for easy analysis.

7. Data Delivery to Base Station

After completing its route, the MuleBot transmitted the collected data to a base station through ESP-NOW or serial communication. The base station stored the final dataset.

8. Testing and Validation

The system was tested in multiple scenarios to ensure correct node detection, successful communication, accurate data collection, and smooth robot movement. Adjustments were made based on observed performance.

Results and Discussion

1. Successful Node Detection

The MuleBot was able to reliably detect all registered sensor nodes within the communication range using ESP-NOW. Detection time was fast and consistent.

2. Accurate Data Collection

Environmental data (temperature, humidity, etc.) was successfully collected from each sensor node. No packet loss or corrupted data was observed during multiple trials.

3. Stable Autonomous Movement

The MuleBot moved smoothly along its predefined path. It maintained communication stability during motion and stopped correctly when a node was detected.

4. Efficient Data Transfer to Base Station

The collected sensor data was successfully sent to the base station and stored in CSV format, allowing easy viewing and analysis.

5. Low Power Consumption

Both MuleBot and sensor nodes operated efficiently with significantly lower power usage compared to Wi-Fi-based solutions.

6. Reliable System Performance

Overall testing showed that the system operated consistently, with successful communication, navigation, and data delivery in every trial.

Table 1. Data received from node 1

	A	B	C	D
1	Data Sensed Timing	Sensor Data		Data Received Time at Base Station
2		Temperature (°C)	Humidity (%)	
3	2025-11-11 22:24:46	25.3	72	2025-11-11 22:25:00
4	2025-11-11 22:24:47	25.3	72	2025-11-11 22:25:00
5	2025-11-11 22:24:48	26.7	69	2025-11-11 22:25:00
6	2025-11-11 22:24:49	26.7	69	2025-11-11 22:25:00
7	2025-11-11 22:24:50	26.7	69	2025-11-11 22:25:00
8				
9				

Node1 (DHT11) data

Table 2. Data received from node 2

	A	B	C
1	Data Sensed Timing	Gas Concentration (ppm)	Data Received Time at Base Station
2	2025-11-11 22:28:25	1312	2025-11-11 22:28:39
3	2025-11-11 22:28:26	1329	2025-11-11 22:28:39
4	2025-11-11 22:28:27	1757	2025-11-11 22:28:39
5	2025-11-11 22:28:28	2918	2025-11-11 22:28:39
6	2025-11-11 22:28:29	2637	2025-11-11 22:28:39
7			
8			

Node2 (MQ6) data

The results of the MuleBot project demonstrate that a mobile data-mule architecture is an effective alternative to traditional wireless communication methods in remote or low-connectivity environments. The use of ESP-NOW significantly improved reliability and reduced power consumption compared to Wi-Fi or long-range modules. This makes the system suitable for battery-powered sensor nodes deployed over large areas.

The MuleBot's autonomous navigation allowed it to move through the field and collect data without manual intervention, proving that mobile collection can overcome the limitations of fixed communication infrastructure. The consistent detection and data transfer from sensor nodes show that short-range communication can still support large sensor networks when paired with mobility.

One important observation is that performance depends on correct node placement and uninterrupted movement of the MuleBot. Obstacles or irregular terrain may affect navigation accuracy, suggesting that future improvements could include better obstacle detection, GPS integration, or advanced path-planning.

Conclusion

The MuleBot project successfully demonstrates a practical and efficient approach to collecting data from distributed sensor nodes using a mobile data-mule architecture. By integrating an ESP32-based robot with ESP-NOW communication, the system reliably detected sensor nodes, gathered their data, and delivered it to the base station with low power consumption and minimal infrastructure requirements.

The results show that this design is cost-effective, energy-efficient, and capable of functioning in environments where continuous wireless connectivity is not feasible. With its stable performance and reliable operation, the MuleBot system can be effectively applied in agriculture, environmental monitoring, remote sensing, and restricted-area surveillance.

Overall, the project proves that mobile data-mule systems offer a strong alternative to traditional fixed communication networks and provide a scalable solution for future IoT deployments.

Future Work

1. Advanced Navigation System

Incorporating GPS, ultrasonic sensors, or computer vision can enable more accurate navigation, obstacle avoidance, and coverage of larger areas.

2. Longer Communication Range

Integrating LoRa or Mesh networking can extend communication distance and allow the MuleBot to collect data from wider and more complex environments.

3. Improved Power Management

Adding solar charging, sleep modes, or low-power path optimization can further increase battery life for both the MuleBot and sensor nodes.

4. Enhanced Data Security

Implementing encryption and authentication mechanisms can strengthen secure data transfer in sensitive applications such as defence or industrial monitoring.

5. Real-Time Dashboard

A cloud or web-based interface can be developed to visualize collected data in real-time at the base station.

6. Additional Sensors and Payloads

The MuleBot can be upgraded to carry more sensors, enabling tasks like soil analysis, gas detection, or surveillance.

7. Multi-MuleBot Coordination

Deploying multiple MuleBots can improve data collection speed and provide redundancy in large-scale sensor networks.

REFERENCES

- 1] F. O. Adunola, V. N. Omeke, and A. A. Ahmad, "Energy-efficient wireless sensor network for real-time environmental monitoring," *Discover Electronics*, 2025.
- [2] D. Joshi and V. Jha, "Energy-efficient multi-static sink model for wireless sensor network," *Journal of Supercomputing*, 2025.
- [3] M. J. Hoque, M. S. Islam, I. Ahmed, and M. Nurullah, "Enhancing Precision Agriculture Efficiency Through Edge Computing-Enabled Wireless Sensor Networks: A Data Aggregation Perspective," *Engineering Proceedings*, 2024.
- [4] O. M. Gul, "Energy-Aware 3D Path Planning by Autonomous Ground Vehicle in Wireless Sensor Networks," *World Electric Vehicle Journal*, 2024.
- [5] M. Abotaleb and J. Mindykowski, "Enhancement of Operational Safety in Marine Cargo Cranes on a Container Ship Through the Application of Authenticated Wi-Fi Based Wireless Data Transmission from Multiple Sensors," *Sensors*, 2024.
- [6] S. K. Erskine, "Secure Data Aggregation Using Authentication and Authorization for Privacy Preservation in WSNs," *Sensors*, 2024.
- [7] W. Khalid and Q. Awais, "Open-Source Internet of Things-Based Supervisory Control and Data Acquisition System for Photovoltaic Monitoring and Control Using HTTP and TCP/IP Protocols," *Energies*, 2024.
- [8] W. He and M. J. A. Baig, "An Open-Source Supervisory Control and Data Acquisition Architecture for Photovoltaic System Monitoring Using ESP32, Banana Pi M4, and Node-RED," *Energies*, 2024.
- [9] O. Tsilomitrou and A. Tzes, "Mobile Data-Mule Optimal Path Planning for Wireless Sensor Networks," *Applied Sciences*, 2022.
- [10] D. Prasannababu and T. Amgoth, "Joint mobile wireless energy transmitter and data collector for rechargeable wireless sensor networks," *Wireless Networks*, 2022.
- [11] A. Khalifeh, K. A. Darabkh, and A. M. Khasawneh, "Wireless Sensor Networks for Smart Cities: Network Design, Implementation and Performance Evaluation," *Electronics*, 2021.
- [12] K. M. S. Srinivasan and L. R., "Design and Development of Wireless Sensor Network based data logger with ESP-NOW protocol," *Convergence in Tech. (6th Int. Conf.)*, 2021

